MILITARY SPECIFICATION

PROPELLANT M5 FOR USE IN CHARGE, PROPELLING, XM178 FOR 90MM RECOILLESS RIFLE

1. SCOPE

1.1 This specification covers M5 Propellant for one type of propelling charge designated as XM178 for the 90MM Recoilless Rifle.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-652 — Propellants Solid for Cannons Requirements and Packing

MIL-STD-1168 — Lot Numbering of Ammunition

DRAWINGS

MILITARY

9214569 — Charge, Propelling, XM178

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "49 CFR 71-90 (latest revision).")

3. REQUIREMENTS

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3.1 Materials. The propellant shall be M5, Type II and conforming to the requirements of Standard MIL–STD–652 and Drawing (dwg) 9214569.

3.2 Ballistics. The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.1.

3.2.1 *Velocity and pressure.* At 70° Fahrenheit (°F) a weight of propellant shall be charge assessed to comply with the prescribed service velocity of 1,100 feet per second (335 meters per second) without exceeding a maximum rated pressure of 7,800 pounds per square inch (psi), piezo equivalent pressure (pep).

3.2.2 *Uniformity of velocity.* The standard deviation in velocity for the lot at 70°F shall not exceed 12 feet per second (3.7 meters per second).

3.2.3 *Permissable individual maximum pressure at 70°F.* No individual maximum pressure shall be greater than 8,600 psi pep (110 percent of the rated maximum pressure of the gun).

3.2.4 *Permissable individual maximum pressure at 145°F.* No individual maximum pressure shall be greater than 8,900 psi, pep.

3.2.5 *Rearward recoil.* The rearward recoil shall be no more than 3 pounds second.

3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of batches of product inspected, (shall be made available upon request by the contracting officer.)

(c) Results obtained, by defect code, for all inspection performed, (shall be made available upon request by the contracting officer).

(d) Drawing, specification number and date, together with an identification and date of changes.

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(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of pounds in the lot.

(g) Date submitted.

(h) Eight copies of propellant description sheet AMC Form 1047 (00 Form 1204) containing the applicable data. (see 6.9).

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawing and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractors records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of one or more batches of M5 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of M5 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.2.2 Testing (see 4.3)

4.2.2.1 *Ballistics (see 3.2).* A composite sample of 100 pounds of the propellant shall be prepared by selecting equal quantities of the propellant from at least 10 randomly selected containers. When lots are comprised of 10 containers or less, each container will be sampled. The prepared sample shall be used for charge establishment, determining uniformity of velocity and determining maximum pressure.

4.2.2.1.1 *Pressure (see 3.2.1) Critical defect, Code No. 01001 (see 6.4).* The lot of propellant shall be rejected if it fails to meet the applicable requirement.

4.2.2.1.2 *Uniformity of velocity (see 3.2.1 and 3.2.2) Major defect, Code No. 02001.* If the product of the facter times the velocity standard deviation of the test rounds exceeds the applicable requirement, the lot shall be rejected. For the initial production lots a 20-round sample will be used and the factor will be 0.79 (see 6.6 and 6.7).

4.2.2.1.3 *Permissable individual maximum pressure (see 3.2.3 and 3.2.4) Critical defect, Code No. 03001.* The maximum pressure calculated for the assessed charge weight for the lot, plus 4 times the standard deviation of the maximum pressure obtained, shall not exceed the applicable requirement (see 6.2).

4.2.2.2 *Rearward recoil, Code No. 04001 (see 3.2.5).* The lot of propellant shall be

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rejected if any round fails to meet the applicable requirement.

4.2.2.3 *Retest.* If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

4.3 Test methods and procedures.

4.3.1 *Ballistics.* These tests shall be conducted at a Government proving ground in accordance with the applicable Ordnance Proof Manual. A weight of charge shall be calculated, from charge velocity and charge-pressure curves representing actual firing data, to give prescribed service velocity.

4.3.1.1 *Uniformity series.* The propellant shall be conditioned to 70°F plus or minus 2°F for at least 16 hours and shall be fired within a minimum time after removal from the conditioning chamber. A 20-round uniformity series shall be fired for evaluation of the initial production lots (see 6.7) and the velocity mean and standard deviation shall be calculated after reducing results to standard conditions and pressure. The pressure shall be reported as piezo equivalent pressure. A 10-round uniformity series shall be fired for all subsequent lots. (Reference shall be made to Dwg 9214569.)

4.3.2 *Conditioned rounds at 145° F.* A 10-round uniformity series shall be fired on rounds which were conditioned for a minimum of 15 days at 145°F before firing. These rounds will be inert loaded except for propellant system. This test will be performed on the first 5 lots which have been accepted. Firings will be performed within a minimum time after removal from the conditioning chamber.

4.3.3 Extreme temperature firings (see 6.8)

4.3.3.1 *Conditioned rounds (minus 65°F).* For informational purposes only, a 10-round uniformity series shall be fired on rounds which were conditioned for a minimum of 24 hours, at minus 65°F before firing. These rounds will be inert loaded except for propellant system. This test will be performed on the first 5 lots which have been accepted. Firings will be performed within a minimum time after removal from the conditioning chamber.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Standard MIL–STD–652.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Packing required (see 5.1)

(c) Process — details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation. The standard deviation shall be calculated with (N–1) as the divisor in a standard statistical technique equivalent to that shown in Standard MIL-STD–414, Section B, Example B–1.

6.3 Submission of test data. In addition to the normal distribution of records, when the propellant is procured by the Department of Army, one copy of all ballistic test data, AMC Form 1060 will be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND2, and one copy to ATTN: SMUPA–DC8, Dover, N.J.

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6.4 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 **Intended use.** It is intended that the item of this specification will be used in the 90MM Recoilless Rifle.

6.6 **Factors 0.79 and 0.73.** The factors 0.79 and 0.73 makes allowances for the probability that a standard deviation from a 20- and 10-round sample respectively exceeds the true standard deviation by chance alone.

6.7 **Initial production lots.** The initial production lots represents the first three consecutive lots which have been produced and accepted.

6.8 **Extreme temperature firings at minus 65° F.** This test will be performed to obtain additional information on the characteristics of the propellant at extreme temperature. Results of the informational testings will have no effect upon the acceptability of the lot(s) involved.

6.9 **Submission of AMC Form 1047.** Information contained in propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.1).

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1375–A967

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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